VigiTrack: AI-Based Unattended Object Detection and Monitoring System Using Computer Vision

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1. System Executive Summary (The 10-Second Pitch)

What VigiTrack Does in Plain English: VigiTrack is an automated AI surveillance decision-support prototype that monitors institutional CCTV video feeds to identify abandoned bags (`backpack`, `handbag`, `suitcase`). Rather than only detecting stationary pixels like basic CCTV cameras do, VigiTrack monitors **human presence alongside the item**. It finds the person closest to the bag, connects them with an invisible proximity leash, and keeps the line **GREEN** while attended. If the person walks away beyond 150 pixels or exits the room, the leash turns **RED** and a 10-second stopwatch begins. If the person returns within 10 seconds, the timer cancels; if they stay away, an alarm sounds and the incident is recorded.

> **ELEVATOR PITCH FOR DEFENSE:** In 2 sentences: "VigiTrack measures the distance between a bag and its nearest custodian. If they stay close, it links them safely; if the person walks away for more than 10 seconds, the stopwatch runs out and alerts security."

2. System Innovation: The Person-Object Association Pipeline

Why Existing Systems Fail (Alarm Fatigue): Conventional CCTV systems (such as built-in tripwire or stationary-item detectors in Hikvision or Dahua NVRs) rely exclusively on stationary object timers. If a student sits on a library bench reading a textbook with their backpack on the floor beside them, conventional systems ring an alarm because the bag is not moving. Because these systems ring non-stop for seated students, security operators turn the feature off completely.

Our Single Primary Contribution: VigiTrack solves this fatal false-alarm problem through a **Context-Aware Person-Object Association Pipeline**. It distinguishes between an item that is **physically attended** (person sitting nearby) versus an item that is **abandoned** (person walked away).

- **Why We Call It "Person Association" (Not "Owner Association"):** From a scientific and legal standpoint, camera pixels cannot verify property ownership or receipt of purchase without invasive biometric profiling. By designating this as **Person-Object Association**, VigiTrack objectively evaluates human custodial proximity without making unverified legal assumptions.
- **Bystander Rejection via Temporal Debouncing:** In naive proximity algorithms, a stranger casually walking past an abandoned bag steals the association and resets the timer. VigiTrack enforces a **5-frame debounce lock** (`DEBOUNCE_COMMIT_FRAMES = 5`), requiring a person to stay within 150 pixels for multiple consecutive frames before locking, while ignoring transient passersby.
- **Frame-Exit Countdown Persistence:** When a person leaves an item and exits the camera's field of view, standard detectors drop the tracking pair. VigiTrack's state machine enters an `Associated Person Out of View` state, keeping the countdown active until security is notified.

3. Mathematical & Algorithmic Implementation

The association engine operates inside `app/detection/fsm_tracker.py` across four deterministic mathematical stages:

Pipeline Stage	Mathematical / Code Logic	Operational Purpose
1. Centroid Calculation	Centroid: `cx = (x1 + x2) / 2`, `cy = (y1 + y2) / 2` for each bag and person bounding box.	Reduces multi-dimensional bounding boxes into single spatial coordinate points in the image plane.
2. Euclidean Distance	Distance `d = sqrt((x_person - x_bag)^2 + (y_person - y_bag)^2)`. Evaluated against `dist_threshold = 150px`.	Identifies the closest human custodian. If `d <= 150px`, the person is tagged as candidate handler.

Pipeline Stage	Mathematical / Code Logic	Operational Purpose
3. 5-Frame Debounce Lock	Candidate handler must maintain `d <= 150px` for 5 consecutive frames before setting `self.owner_id = person_id`.	Bystander protection: prevents random pedestrians walking past from resetting or stealing association.
4. 6-State FSM Monitoring	States: `UNASSIGNED -> CARRIED -> STATIC_ATTENDED -> SEPARATED -> OUT_OF_VIEW -> UNATTENDED_ALERT`.	Controls timer lifecycle: resets stopwatch if person returns within 10s; sounds alarm if separation exceeds 10s.

4. Dissecting the Architecture: What is AI vs. What is Rules?

A frequent trap in capstone defenses is claiming the entire system is "AI." Knowing the exact technical boundary demonstrates high engineering maturity:

Subsystem Component	Is it AI?	Underlying Technology & Mechanism	Where in Code?
Object Detection	**YES (AI)**	Convolutional Neural Network (CNN) trained on MS COCO dataset to extract visual features and classify items.	`app/detection/detector.py` (`YOLOv8m` / `YOLOv8n`)
Multi-Object Tracking	**YES (AI)**	ByteTrack: Kalman Filter motion forecasting combined with Hungarian Algorithm IoU association across video frames.	`app/detection/detector.py` (`tracker="bytetrack.yaml"`)
Distance Calculation	**NO (Math)**	Standard Pythagorean / Euclidean distance formula measuring straight-line pixel gaps.	`app/detection/fsm_tracker.py` (`math.sqrt()`)
Abandonment Timer	**NO (Rules)**	Deterministic Finite State Machine checking elapsed clock delta (`time.time() - start_time > 10.0s`).	`app/detection/fsm_tracker.py` (`BaggageTracklet`)
Overlay Drawing & Stream	**NO (Graphics)**	OpenCV matrix manipulation drawing green/red lines and encoding MJPEG byte buffers.	`app/stream/camera_manager.py` (`cv2.line`, `cv2.imencode`)
Web UI & Database	**NO (Web/DB)**	Flask HTTP routing, Jinja2 templates, and SQLite relational ACID storage.	`app/stream/routes.py` `app/database/models.py`

5. Integrating with Live CCTV Infrastructure (Sir Serge's Questions)

1. Plugin vs. Standalone Application: VigiTrack operates as a **standalone Edge Analytics Server**. It does not install as a proprietary plugin inside NVR firmware (such as Hikvision iVMS or Dahua SmartPSS), nor does it flash custom code onto camera chips. It runs independently on a local workstation connected to the security network.

2. How Live Video Streams into the Application (RTSP Protocol): Institutional CCTV cameras and NVRs universally broadcast video over the Local Area Network (LAN) using **RTSP (Real-Time Streaming Protocol)** on standard port **554**. VigiTrack connects to the NVR's video feed using OpenCV's FFmpeg backend:

```
cap = cv2.VideoCapture("rtsp://admin:SecurityPass@192.168.1.100:554/Streaming/Channels/102")
```

3. Dual-Stream Strategy (Main-Stream vs. Sub-Stream): The NVR records the high-bitrate **Main-Stream** (1080p/4K) to hard drives for legal evidence. VigiTrack pulls the lightweight **Sub-Stream** (720p/D1 at ~15–30 FPS). Because YOLOv8 resizes incoming frames to `640x640` anyway, using the sub-stream cuts network decoding load by 60% and guarantees real-time 30 FPS processing without saturating campus bandwidth.

4. Physical Deployment Architecture (The 3 Real-World Rooms):

- **Room 1: The Hallway / Ceiling:** The school's existing IP cameras and PoE switches, untouched by VigiTrack.
- **Room 2: The Server Room / Tech Desk:** The standalone PC running VigiTrack, pulling RTSP streams, executing AI inference, and updating SQLite.
- **Room 3: The Security Guard Booth:** A standard guard monitor opening Google Chrome or Edge to `http://127.0.0.1:5000` with audio speakers for alarms. The guard requires zero technical setup.

6. Hardware Requirements: Two-Tier Architecture

Does VigiTrack Require an NVIDIA RTX GPU? No. VigiTrack features a **Two-Tier Hardware Architecture** that allows institutions to deploy the system without purchasing expensive gaming or workstation graphics cards:

- **Tier 1: High-Compute Dedicated GPU Tier (NVIDIA CUDA):** When deployed on a dedicated security server with a discrete graphics card, `app/detection/detector.py` automatically detects CUDA and loads **YOLOv8 Medium (`yolov8m.pt`)** for maximum detection precision on small bags across wide-angle lenses.
- **Tier 2: Standard Office Commodity Tier (Intel Core i5 / CPU):** When deployed on standard school office computers with integrated graphics, the system automatically switches to **YOLOv8 Nano (`yolov8n`) compiled with Intel OpenVINO**. Intel OpenVINO accelerates inference across both CPU cores and the on-board Intel Integrated Graphics (iGPU), running at **25–30 FPS** with low CPU utilization.

7. Deployment Justification: Why an Edge / Local Server is Best

When panelists ask: *"Why not host VigiTrack on cloud servers like AWS or Google Cloud?"*, provide these four technical justifications:

- **1. Internet Bandwidth Saturation:** A single 24/7 CCTV camera consumes 2–4 Mbps of continuous upload bandwidth (~600 GB to 1 TB per month per camera). Uploading multiple video feeds will choke campus internet.
- **2. Recurring Cloud GPU Costs:** Running a continuous cloud GPU instance (e.g., AWS EC2 `g4dn.xlarge` with an NVIDIA T4) costs **$150 to $350+ USD every single month**, whereas a local PC is a one-time purchase.
- **3. Operational Reliability During Outages:** If the school loses internet connection during a storm, a cloud surveillance system completely dies. An on-premise edge server on the local LAN continues monitoring 24/7.
- **4. Data Privacy Compliance (RA 10173):** Under the Philippine Data Privacy Act of 2012 and NPC Circular 2024-02, streaming recognizable video of students and minors across public cloud infrastructure creates severe compliance risks. Processing video locally on an air-gapped security network preserves privacy.

8. Deep-Dive: Database, Streaming, Audio & Settings

1. SQLite Database & Incident Deduplication (`app/database/models.py`): Stores structured incident logs in `data/vigitrack.db` (`object_class`, `object_track_id`, `person_track_id`, `distance`, `camera_name`, `detected_at`). To prevent flooding the database with 30 duplicate records per second while a bag sits unattended, `camera_manager.py` uses an in-memory set (`self.logged_ids`) ensuring each incident is recorded **exactly once** per abandonment event.

2. HTTP Motion-JPEG Video Streaming Engine (`app/stream/routes.py`): Instead of requiring complex WebRTC media servers or browser plugins, VigiTrack uses **MJPEG over HTTP** (`multipart/x-mixed-replace; boundary=frame`). OpenCV compresses each annotated frame into a 960p JPEG (`quality=75`), and the browser renders it inside a native `<img src="/video_feed">` tag like a continuous, zero-latency flipbook.

3. Threaded Audio Alarm & Visual HUD (`app/alerts/notifier.py`): Playing audio tones via `winsound.Beep()` blocks execution for 400ms. VigiTrack dispatches sound playback in a **dedicated background thread** so the video feed never stutters while the siren sounds. A semi-transparent red HUD banner (`cv2.addWeighted`) provides instant visual confirmation.

4. Persistent Settings Store (`config/settings_store.py`): Operator threshold calibrations (`distance_threshold`, `time_threshold`, class filters) are saved to `data/settings.json`. They hot-reload into memory instantly without stopping the camera feed and persist across PC reboots, with automatic fallback to safe defaults if corrupted.

5. Thread Synchronization (`threading.Lock`): A camera reader thread ingests frames while Flask request threads read frames for streaming. Thread locks (`with self.lock:`) prevent race conditions, memory corruption, and partial image tearing.

9. Master Defense Script & FAQ Quick Reference

Panelist Question	Recommended Response / Defense Script
"Why use your system instead of existing CCTV features?"	Existing systems only monitor stationary pixels and trigger false alarms whenever students sit beside their bags. VigiTrack introduces **Person-Object Association**, alerting only when the human custodian actually separates and walks away.
"How do you know that person actually owns the bag?"	We specifically term this **Person-Object Association** rather than legal ownership. In CCTV surveillance, camera pixels cannot verify legal ownership; our system tracks **custodial human proximity** as an objective decision-support metric.
"What happens if a bystander walks past the abandoned bag?"	Our **5-frame debounce lock** requires 5 consecutive frames of proximity before binding. Random passersby walking in front of the bag are rejected and cannot steal association or reset the countdown.

Panelist Question	Recommended Response / Defense Script
"What happens if the person walks out of the camera view?"	The system enters `Associated Person Out of View`. The bounding line disappears, but the 10-second countdown continues uninterrupted until the security alert is dispatched.
"Does the school need to buy an expensive graphics card?"	No. While developed on a dedicated GPU tier, VigiTrack includes an **Intel OpenVINO-optimized YOLOv8 Nano** mode that runs at real-time 25–30 FPS on standard Core i5 school office desktop computers.
"Why not run this in the cloud?"	Streaming 24/7 CCTV video over the internet chokes school bandwidth, costs hundreds of dollars monthly in cloud GPU fees, fails during internet outages, and violates student data privacy (RA 10173). Edge processing is the surveillance industry standard.